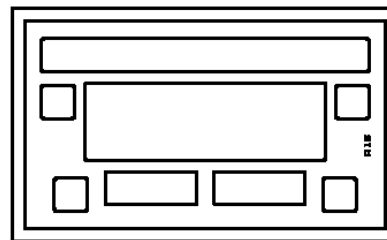


GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

FEATURES

- Operating Frequency Range: DC to 7GHz
- Operating Drain Voltage: 28V & 50V
- Maximum Output Power (P_{SAT}): 30W
- Suitable for CW, Pulsed, Linear applications
- 100% KGD DC Production Tested



1.265 x 0.755 mm Die

DESCRIPTION

The GRF0020D is a 30W (P3dB) unmatched discrete GaN-on-SiC HEMT which operates from DC to 7GHz on a 50V supply rail. The wide bandwidth of the GRF0020D makes it suitable for a variety of applications including cellular infrastructure, radar, communications, and test instrumentation, and can support CW, linear and pulse operations.

Bare die are shipped in 2-inch waffle trays for safe transport and storage.

Typical Performances Measured Loadpull 1 Tone pulsed CW (10% duty cycle, 100 μ s width) in DFN 3x3 package, *2nd Harmonics NOT optimized*

- (1) Optimum Peak Power at 2.5dB in compression
- (2) Optimum Peak Efficiency at 2.5dB in compression

V_{ds}=50V, I_{dq} = 30mA, T_A = 25°C

Frequency (MHz)	P _{out} ⁽¹⁾ (dBm)	Gain ⁽²⁾ (dB)	Eff ⁽²⁾ (%)
1000	45	24.3	65.7
1400	44.9	22.1	66.9
1800	45.1	19.8	63.7
2200	45.1	19.6	60.7
2600	45.2	18.8	64.3
3000	45.4	18.7	63.8
4000	45.4	17.2	65
5000	45.1	15.1	60
5500	45.2	14.2	60.6
6000	45.3	13.8	61.6

V_{ds}=28V, I_{dq} = 30mA, T_A = 25°C

Frequency (MHz)	P _{out} ⁽¹⁾ (dBm)	Gain ⁽²⁾ (dB)	Eff ⁽²⁾ (%)
1000	42.2	20.5	69.1
1400	42.3	19.5	69.1
1800	42.6	18.4	67.5
2200	42.6	17.7	61.7
2600	42.8	17.4	65.7
3000	42.8	16.8	65.8
4000	42.7	14.9	65.9
5000	42.5	12.6	61.5
5500	42.5	12	62.7
6000	42.6	11.8	63.6

GRF0020D

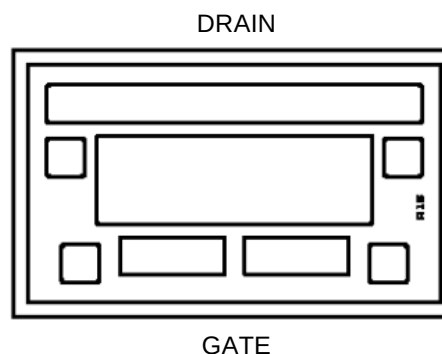
50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

ABSOLUTE MAXIMUM RATINGS^(1, 2)

Parameter	Rating	Symbols and Units
Drain Source Voltage	150	V_{DS} (V)
Gate Source Voltage	-8 to +2	V_{GS} (V)
Operating Voltage	55	V_{dsq} (V)
Junction Temperature	+225	T_{JUNC} (°C)
Storage Temperature	-65 to +150	$T_{STORAGE}$ (°C)

- Exceeding any of these limits may cause permanent damage to this device or seriously limit the lifetime (MTTF).
- Guerrilla RF does not recommend sustained operation above maximum operating conditions.

BLOCK DIAGRAM

ELECTRICAL SPECIFICATIONS: $T_A = 25^\circ\text{C}$

Parameter	Min.	Typ.	Max.	Symbols and Units	Test conditions
Frequency Range	DC		7000	MHz	
DC Characteristics					
Drain Source Breakdown Voltage	150			V_{BDSS} (V)	
Drain Source Leakage Current		1.27		I_{DLK} (mA)	$V_{gs} = -8V, V_{ds} = 50V$
Gate Threshold Voltage	-3.4		-1.5	V_{GS} (V)	$V_{ds} = 50V$
Operating Conditions					
Gate Bias Voltage		-2.5		V_{GSQ} (V)	
Drain Voltage		50		V_{DSQ} (V)	
Quiescent Drain Current		31		I_{DQ} (mA)	

GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

THERMAL AND RELIABILITY INFORMATION -CW ^(1, 2, 3): T_c = 85°C

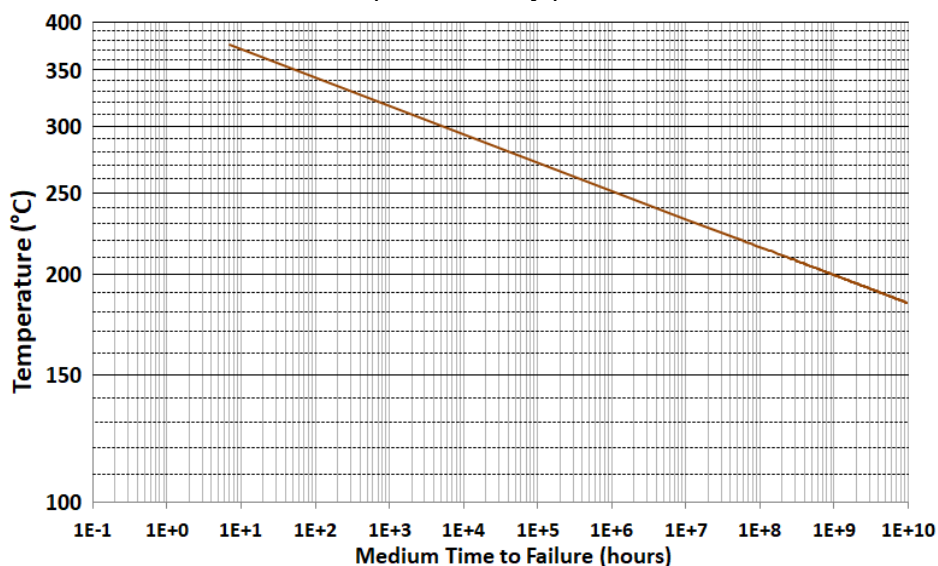
$$R_{th}(^{\circ}\text{C/W}) = 0.0755 \times P_{diss}(\text{W}) + 3.3824$$

Parameter	Test condition	Value	Units
Channel Temperature, T _{ch}	P _{diss} 6 W	124.1	°C
R _{th}		3.9	°C/W
R _{sur}		2.35	°C/W
MTTF		>1.0E+10	Hrs
Channel Temperature, T _{ch}	P _{diss} 12 W	168	°C
R _{th}		4.2	°C/W
R _{sur}		2.64	°C/W
MTTF		>1.0E+10	Hrs
Channel Temperature, T _{ch}	P _{diss} 18 W	220	°C
R _{th}		4.8	°C/W
R _{sur}		2.83	°C/W
MTTF		6.0E+07	Hrs

1.Assumes eutectic attach using 1mil low temp solder, mounted to a 8 mil DFN package.

2.Thermal Resistance using Finite Element Analysis (FEA) simulation, calibrated with Infrared measurement on surface temperature.

3.R_{sur}: Thermal resistance based on Surface Temperature, only provided as a reference.



GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

LOADPULL MEASUREMENT, $V_{ds} = 50V$ $I_{dq} = 30mA$

Die packaged in DFN 3x3. Measured 1 Tone Pulse CW, pulse width 100us, duty cycle 10%

For Optimum Peak Power @ 2.5dB Compression

Freq-MHz	Zin_F0	ZI_F0	Gain-dB	Pout-dBm	Pout-W	Eff-%	AMPM-deg
1000	3.7 - j 17.8	26.1 + j 11.4	23.7	45	31.3	61.1	-0.2
1200	4.8 - j 16.0	23.5 + j 8.0	21.6	44.8	30.5	58	1.8
1400	4.1 - j 13.3	22.5 + j 9.0	21.2	44.9	31.2	56.8	1.7
1600	4.0 - j 10.9	19.0 + j 8.2	20.4	45.3	34.1	55	1.2
1800	2.9 - j 8.6	19.3 + j 11.0	19.6	45.1	32.3	55.7	2.3
2000	3.7 - j 7.2	25.5 + j 8.6	17.5	44.8	30.9	51	4.8
2200	2.3 - j 6.0	14.4 + j 7.4	18.3	45.1	32.1	48.8	2.6
2400	2.2 - j 4.5	14.2 + j 9.8	19.3	45	31.3	51.6	1.4
2800	2.1 - j 2.0	15.4 + j 7.2	17	45.4	34.8	52.8	2.6
3000	1.9 - j 0.8	16.2 + j 5.4	17	45.4	34.4	51.4	1
3500	1.8 + j 2.0	10.6 + j 5.3	16.5	45.3	34	52.6	-0.5
4000	2.1 + j 4.7	10.2 + j 4.8	16.1	45.4	34.5	55.8	-1.3
4500	2.1 + j 7.6	8.5 + j 4.4	14.8	45.2	33.4	55.9	-1.6
5000	2.6 + j 10.3	9.5 + j 0.4	13.7	45.1	32.8	51.7	-1.1
5500	3.0 + j 12.6	8.3 - j 1.1	13	45.2	33	53.2	-1.9
6000	3.8 + j 17.8	8.6 - j 3.4	12.4	45.3	34.3	54.8	-1.8

For Optimum Peak Efficiency @ 2.5dB Compression

Freq-MHz	Zin_F0	ZI_F0	Gain-dB	Pout-dBm	Pout-W	Eff-%	AMPM-deg
1000	3.3 - j 15.5	33.9 + j 25.7	24.3	43.6	23.4	65.7	-1.4
1200	3.7 - j 13.2	31.7 + j 26.2	22.5	43.7	23.4	64.5	0.6
1400	3.4 - j 11.9	25.1 + j 19.8	22.1	44.6	29	66.9	1.3
1600	3.8 - j 8.7	28.2 + j 27.6	19.9	44	25.1	66.4	1.7
1800	3.0 - j 6.7	23.9 + j 25.9	19.8	44	25.2	63.7	4.8
2000	3.1 - j 6.2	20.6 + j 22.5	19.2	44.2	26.2	60.4	4.3
2200	2.2 - j 4.5	14.4 + j 20.9	19.6	44	25.3	60.7	3.2
2400	2.0 - j 3.3	13.5 + j 19.8	19.3	44	25.3	59.2	4.3
2800	1.7 - j 0.7	12.5 + j 17.1	18.9	44.6	29.3	63.2	2.4
3000	1.5 + j 0.7	11.2 + j 17.3	18.7	44.2	26.3	63.8	1.5
3500	1.5 + j 3.3	6.8 + j 10.9	17.7	44.3	27.3	62.6	-0.4
4000	1.6 + j 5.9	6.3 + j 9.5	17.2	44.1	25.8	65	-1
4500	1.9 + j 8.1	5.1 + j 6.2	15.8	44.2	26.3	61.2	-2.1
5000	1.9 + j 11.6	5.1 + j 3.7	15.1	44	25.4	60	-2
5500	2.4 + j 13.8	4.5 + j 0.8	14.2	44.2	26	60.6	-3.1
6000	3.0 + j 19.1	5.1 - j 1.5	13.8	44.5	28.1	61.6	-2.6

GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

LOADPULL MEASUREMENT, $V_{ds} = 28V$ $I_{dq} = 30mA$

Die packaged in DFN 3x3. Measured 1 Tone Pulse CW, pulse width 100us, duty cycle 10%

For Optimum Peak Power @ 2.5dB Compression

Freq-MHz	Zin_F0	Zl_F0	Gain-dB	Pout-dBm	Pout-W	Eff-%	AMPM-deg
1000	4.7 - j 18.2	18.3 + j 1.6	21.2	42.2	16.8	63.2	-0.3
1200	4.0 - j 16.0	13.9 + j 0.9	19.6	42.1	16.4	54.8	1.9
1400	3.5 - j 12.6	16.9 + j 4.2	19.7	42.3	17.3	63.5	1.1
1600	3.2 - j 9.9	13.5 + j 2.8	18.4	42.6	18.2	59.3	-1.3
1800	3.1 - j 8.5	15.7 + j 3.6	18.4	42.6	18.3	58.8	1.3
2000	3.1 - j 7.1	17.2 + j 2.7	17.2	42.4	17.3	54.4	2.9
2200	3.0 - j 5.6	16.5 + j 1.5	16.9	42.6	18.1	55.3	2
2400	2.7 - j 4.4	16.5 + j 1.5	16.4	42.4	17.5	53.3	2.5
2800	2.2 - j 1.8	13.5 + j 1.7	16.7	42.9	19.2	56.9	0.3
3000	2.2 - j 0.6	16.0 + j 0.9	15.7	42.8	19	57.5	0.1
3500	2.0 + j 2.1	10.9 + j 1.3	15.4	42.9	19.4	58.5	-1.7
4000	1.8 + j 5.1	11.3 - j 0.4	14.3	42.7	18.9	58.2	-1.6
4500	2.2 + j 7.5	10.0 - j 1.0	13.2	42.6	18.5	57.4	-2.3
5000	2.2 + j 10.7	9.8 - j 4.2	12.4	42.5	18.1	55.6	-2.2
5500	2.6 + j 13.1	9.9 - j 6.9	11.3	42.5	18.2	53.6	-2.5
6000	3.2 + j 18.8	10.1 - j 7.5	11	42.6	18.7	58.4	-2.3

For Optimum Peak Efficiency @ 2.5dB Compression

Freq-MHz	Zin_F0	Zl_F0	Gain-dB	Pout-dBm	Pout-W	Eff-%	AMPM-deg
1000	4.6 - j 16.5	30.3 + j 3.7	20.5	41.4	14.1	69.1	2.1
1200	4.3 - j 13.4	29.8 + j 8.2	19.7	41.1	12.8	67.7	3.8
1400	3.3 - j 10.9	25.3 + j 13.4	19.5	41.2	13.2	69.1	2.8
1600	3.6 - j 8.1	20.7 + j 11.6	18.4	41.4	14	69.6	-1.9
1800	2.7 - j 6.0	17.9 + j 16.3	18.4	40.8	12.3	67.5	1.5
2000	2.1 - j 4.9	16.0 + j 16.2	18	40.6	11.7	59.5	3.2
2200	1.9 - j 4.2	16.2 + j 10.0	17.7	41.9	15.9	61.7	2.5
2400	2.0 - j 3.5	14.8 + j 10.9	17.6	41.7	14.8	60.6	2.6
2800	1.5 + j 0.1	11.8 + j 11.9	17.2	41.5	14.1	65.5	0.8
3000	1.5 + j 1.1	12.0 + j 11.6	16.8	41.3	13.4	65.8	0.1
3500	1.6 + j 3.4	8.7 + j 6.4	15.3	42	16.1	66.1	-2
4000	1.6 + j 6.3	8.2 + j 5.0	14.9	41.6	14.9	65.9	-2.6
4500	1.8 + j 9.2	5.4 + j 3.0	13.9	41	12.7	64.2	-4.6
5000	2.1 + j 11.9	7.0 + j 1.1	12.6	41.3	13.7	61.5	-2.7
5500	2.3 + j 14.5	5.4 - j 2.6	12	41.3	13.6	62.7	-5.9
6000	2.9 + j 20.0	5.7 - j 5.0	11.8	41.4	13.9	63.6	-4.8

GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

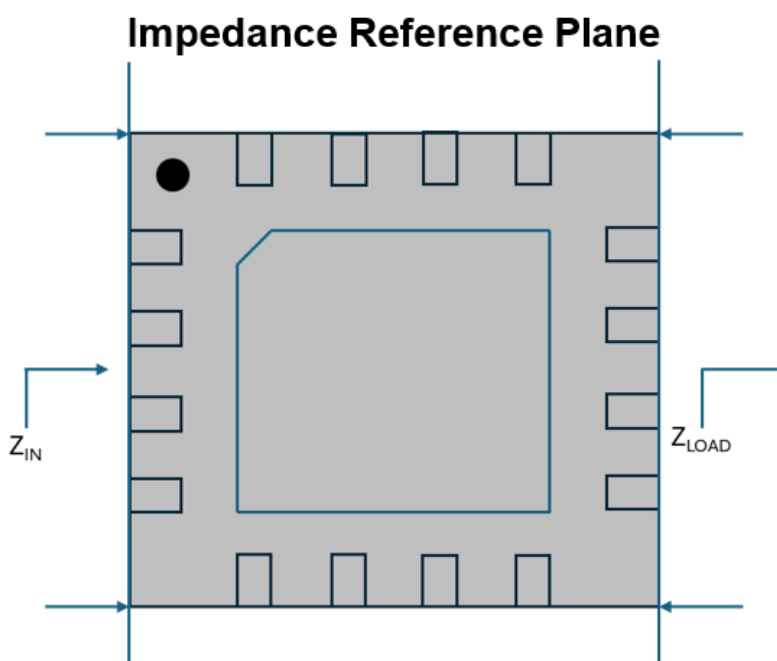
LOADPULL MEASUREMENT NOTES

Source and Load impedance at 2nd Harmonic are set to 10 Ohms.

With proper 2nd Harmonic termination, expect +5% Efficiency for Source and similar with Drain 2nd Harmonic.

Z_{LOAD} : Measured Impedance presented to the Drain of the device.

Z_{IN} : Measured input Impedance at the Gate of the device.



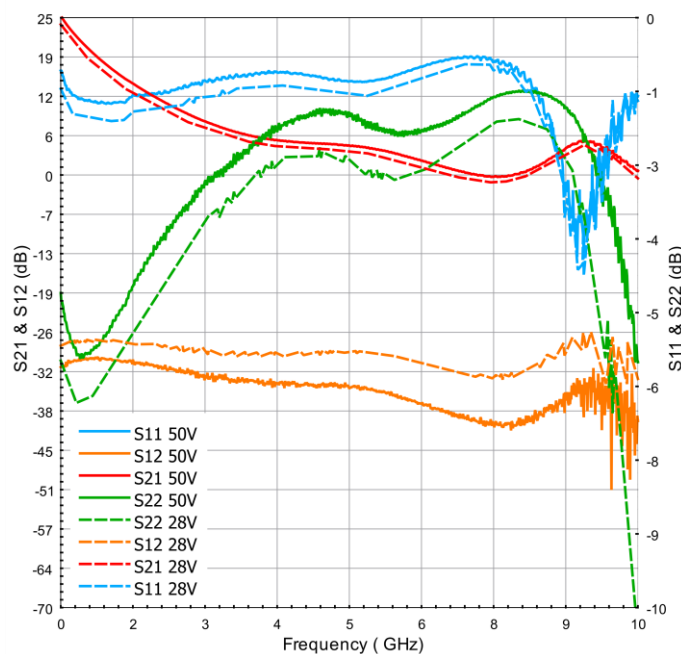
Raw data and full Loadpull measurement report available at request: sales@galliumsemi.com

GRF0020D

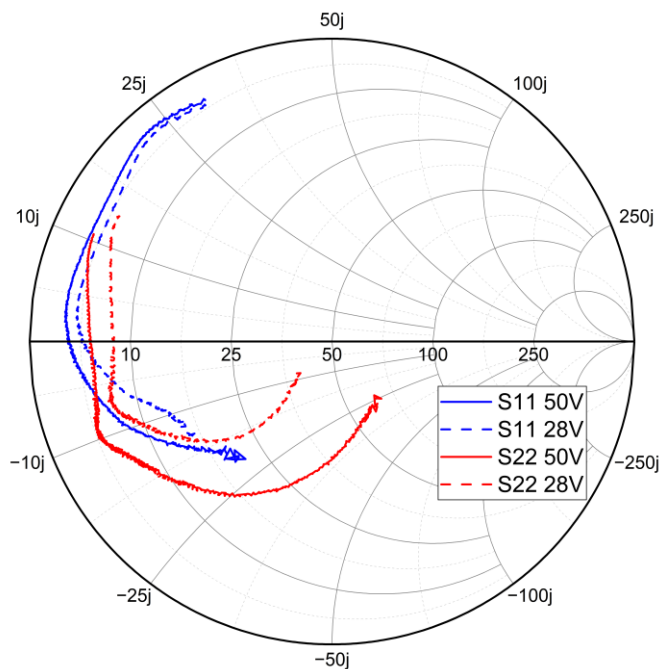
50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

BROADBAND S-PARAMETERS MEASUREMENT, $V_{ds} = 28$ & $50V$ $I_{dq} = 30mA$
 Die packaged in DFN 3x3. Measured 1 Tone CW

S Parameters (Mag-dB)



S11 & S22 0.4-8 GHz

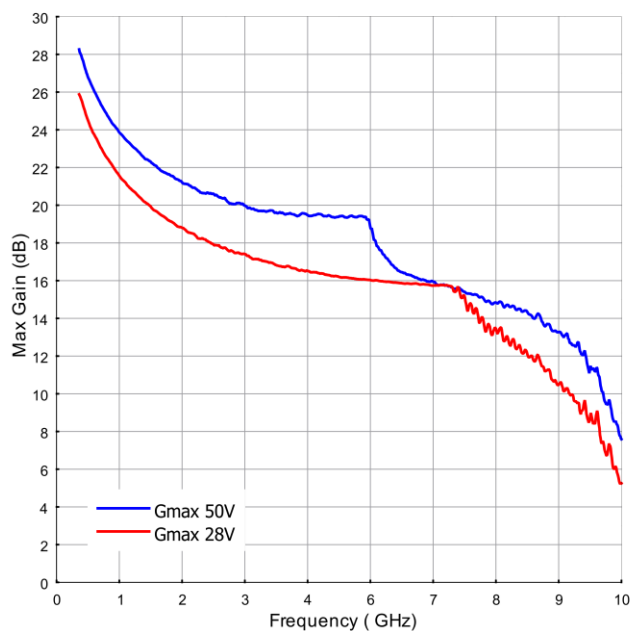


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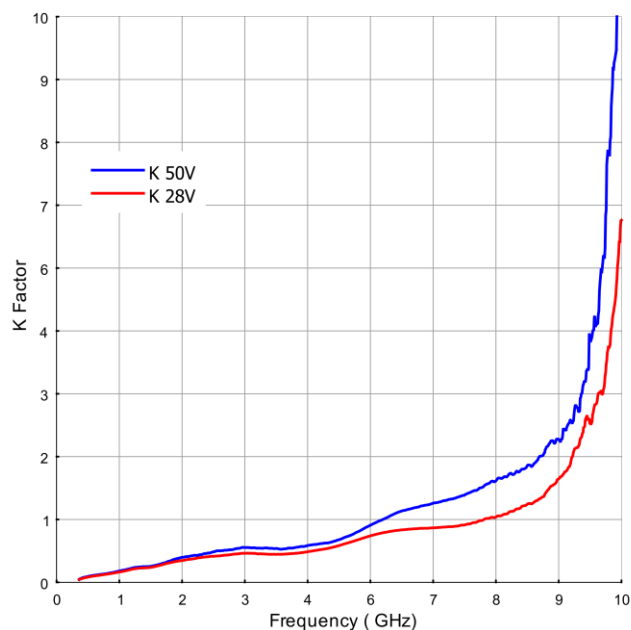
50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

BROADBAND S-PARAMETERS MEASUREMENT, $V_{ds} = 28$ & $50V$ $I_{dq} = 30mA$
Die packaged in DFN 3x3. Measured 1 Tone CW

Maximum Available Gain



K Factor



**GRF0020D****50V, DC – 7GHz, 20W GaN HEMT, BARE DIE**

GaN HEMT BIASING SEQUENCE

To turn the transistor ON

1. Set V_{GS} to -5V.
2. Turn on V_{DS} to normal operation voltage (50V).
3. Slowly increase V_{GS} to set I_{DQ} current (30mA).
4. Apply RF power.

To turn the transistor OFF

1. Turn the RF power off.
2. Decrease V_{GS} to -5V.
3. Turn off V_{DS} . Wait a few seconds for drain capacitor to discharge.
4. Turn off V_{GS} .

GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

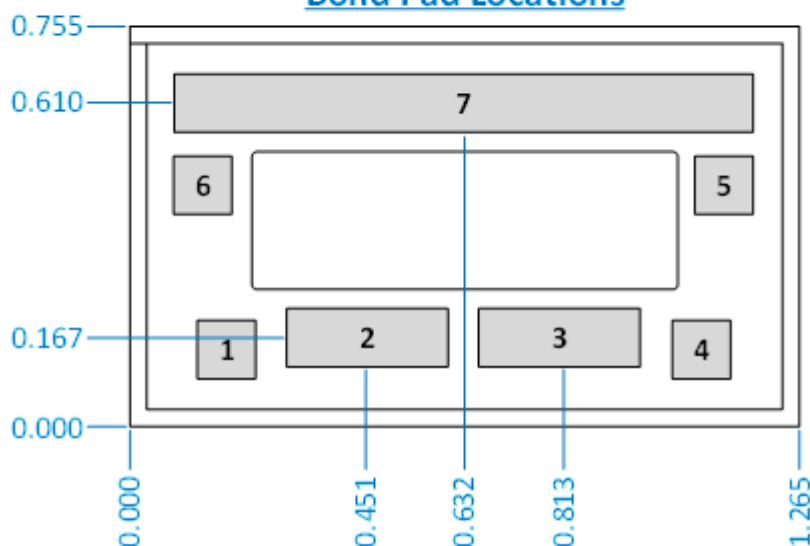
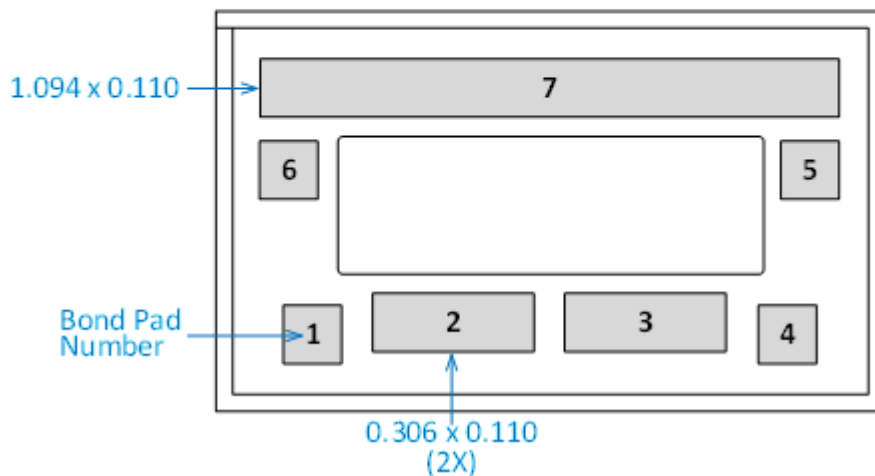
DIE DIMENSIONS

Notes:

- All dimensions are in mm
- Die size is 1.265 X 0.755mm
- Die thickness is 75 μ m
- Bond pad metal is gold
- Backside metal is gold

Bond Pad IDs:

- 1: No Connect
- 2: Gate
- 3: Gate
- 4: No Connect
- 5: No Connect
- 6: No Connect
- 7: Drain

Bond Pad Locations**Bond Pad Sizes**

GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

HANDLING PRECAUTIONS

Parameter	Symbol	Class	Test Methodology
ESD* – Human Body Model	HBM	Class 1A (250V)	ANSI/ESDA/JEDEC Standard JS-001
ESD* – Charged Device Model	CDM	Class C2A (500V)	ANSI/ESDA/JEDEC Standard JS-002

* Tested in DFN 3x3 package



Guerrilla RF Policy on EU RoHS available online:

https://www.guerrilla-rf.com/files/ugd/3748d3_1107b9788f9845f78f45d424097c4c97.pdf



GRF0020D

50V, DC – 7GHz, 20W GaN HEMT, BARE DIE

REVISION HISTORY

Revision Date	Description of Change
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11-14-2024	Release 0 Data Sheet. Added new Z reference plane & die dimension drawings.
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CONTACT INFORMATION

To request the latest information and samples, please contact us at:

Web: www.guerrilla-rf.com • (336) 510-7840

Email: sales@guerrilla-rf.com • 2000 Pisgah Church Road. Greensboro, NC 27455

Data Sheet Classifications

Data Sheet Status	Notes
Advance	S-Parameter and NF data based on EM simulations for the fully packaged device using foundry-supplied transistor S-Parameters. Linearity estimates based on device size, bias condition and experience with related devices.
Preliminary	All data based on limited evaluation board measurements taken within the Guerrilla RF Applications Lab. All parametric values are subject to change pending the collection of additional data.
Release 0	All data based on measurements taken with <i>production-released</i> material. TYP values are based on a combination of ATE and bench-level measurements, with MIN/MAX limits defined using <i>modelled estimates</i> that account for part-to-part variations and expected process spreads. Although unlikely, future refinements to the TYP/MIN/MAX values may be in order as multiple lots are processed through the factory.
Release A-Z	All data based on measurements taken with production-released material <i>derived from multiple lots which have been fabricated over an extended period of time</i> . MIN/MAX limits may be refined over previous releases as more statistically significant data is collected to account for process spreads.

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**GRF0020D****50V, DC – 7GHz, 20W GaN HEMT, BARE DIE**

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